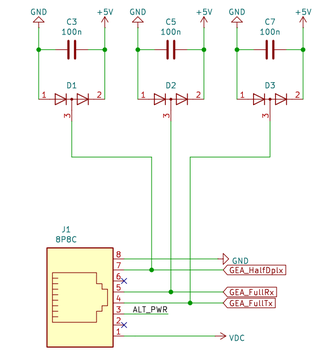
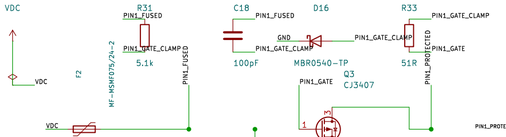
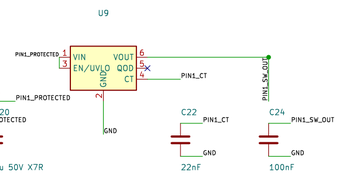
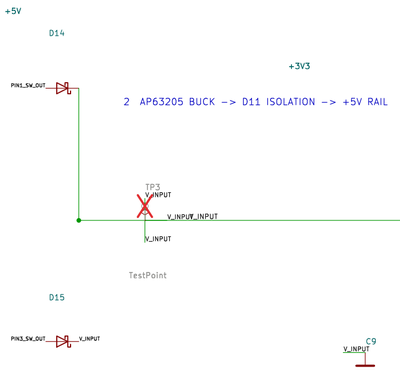
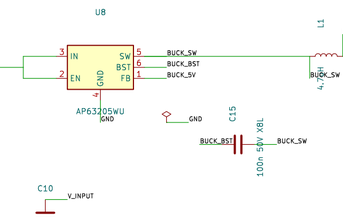
1 APPLIANCE CONNECTOR / PROTECTED INPUTS

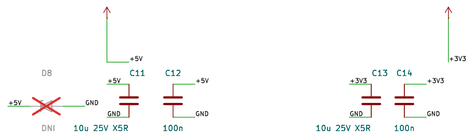


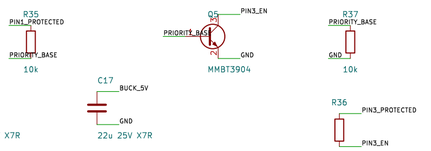


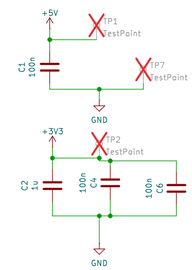




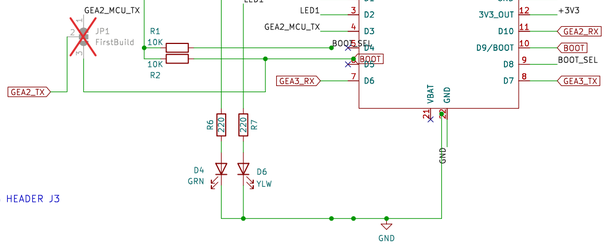


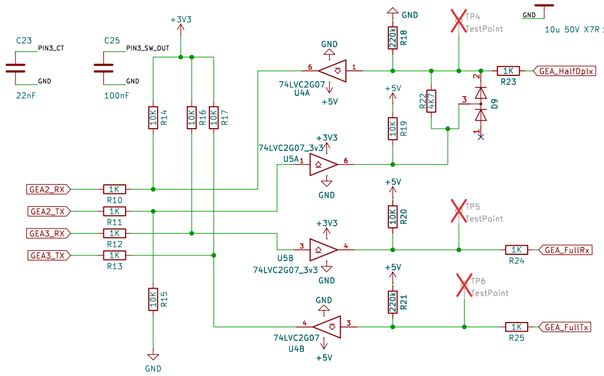




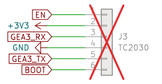


4 XIAO ESP32-C3 MODULE / UART / LEDS





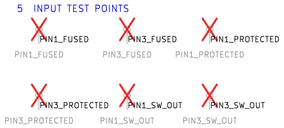






XIAO ESP32-C3 (22-pad OPL symbol). Native USB stays on the module.
Pad 14 VUSB is tied directly to +5V; pad 21 VBAT is unused.

5 INPUT TEST POINTS



7 PROTECTED-INPUT TVS CLAMP



6 RECOVERY HEADER J2 (5V IN VIA D19) / DEBUG HEADER J3

POWER AND OPERATING NOTES

Appliance pin 1 / pin 3 selection is automatic (U9/U10 load switches, D14/D15 blocking); D1/D2/D3 are input ESD clamps.
U8 AP63205 buck output BUCK_5V reaches the +5V rail through D11, which blocks backfeed into the buck.
XIAO pad 14 VUSB is the module native USB VBUS net and sits directly on +5V; the only fuse is inside the module, downstream of pad 14. XIAO pad 21 VBAT is unused.
J2 is a permanent recovery header: regulated 5V in through D19, 3.3V logic. Use it only with USB and the appliance physically disconnected.
CONNECT ONLY ONE SUPPLY AT A TIME. With the appliance powered the USB connector VBUS is energised; simultaneous sources are not protected.
U9/U10 QOD pin 5: NC. DNP = do not populate: test points and mounting holes are not assembled components.

UNQUALIFIED PROTOTYPE – REVIEW BEFORE FABRICATION
Dr-Crow
Sheet: /
File: GEA-Adapter-Rev3B.kicad_sch

Title: GE Appliance Adapter – Rev3B XIAO integration

Size: A2	Date: 2026-09-28	**Rev: 3B**
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